Infratemporal fossa & TMJ

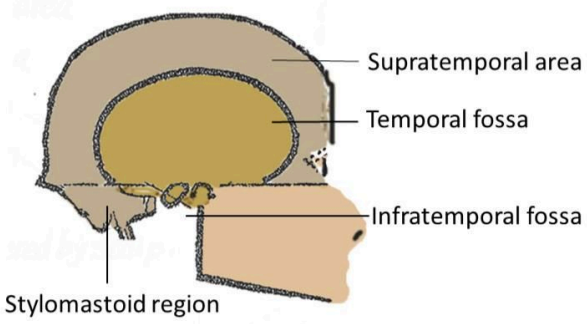
The space behind the maxilla and under the skull base: boundaries, muscles of mastication, maxillary artery, pterygoid plexus, mandibular nerve and the TMJ.

1. Boundaries and contents
2. Muscles of mastication
3. Maxillary artery
4. Pterygoid venous plexus
5. Mandibular nerve in the fossa
6. Temporomandibular joint
7. On imaging
8. In theatre

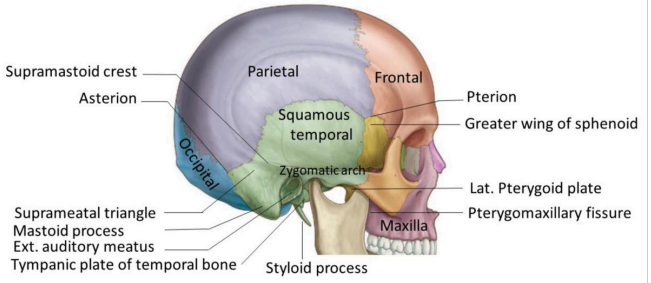
01 Boundaries and contents

- **Roof:** infratemporal surface of the greater wing of sphenoid (with **foramen ovale** and **foramen spinosum**).
- **Lateral:** ramus of the mandible. **Medial:** lateral pterygoid plate. **Anterior:** back of the maxilla. **Posterior:** styloid process and carotid sheath. Open below.
- **Communicates** with the temporal fossa (deep to the zygomatic arch), the **pterygopalatine fossa** (pterygomaxillary fissure), the orbit (inferior orbital fissure) and the middle cranial fossa (foramina ovale and spinosum).
- **Contents:** lateral and medial pterygoid muscles, sphenomandibular ligament, **1st and 2nd parts of the maxillary artery**, **pterygoid venous plexus**, **mandibular nerve** and branches, **chorda tympani**, **otic ganglion**.

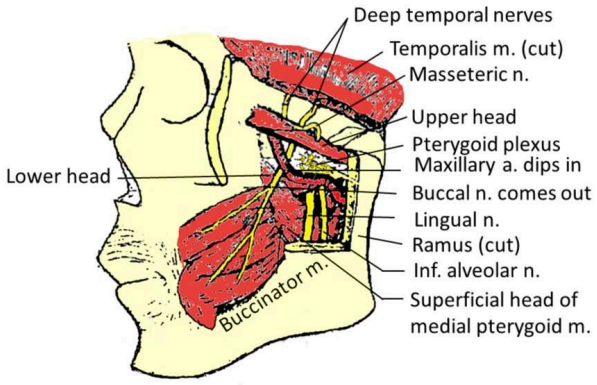
> **CLINICAL PEARL** Tumours of the parapharyngeal space, nasopharynx and maxilla reach the infratemporal fossa; trismus means the pterygoid muscles are involved.



Temporal and infratemporal fossae
Dr. Sameh Doss handout, p. 8

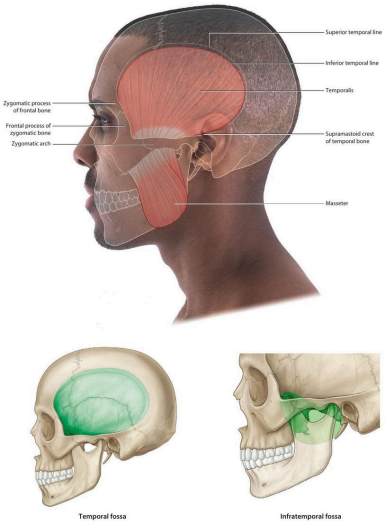


Lateral skull: pterion, asterion and the infratemporal surface
Dr. Sameh Doss handout, p. 9



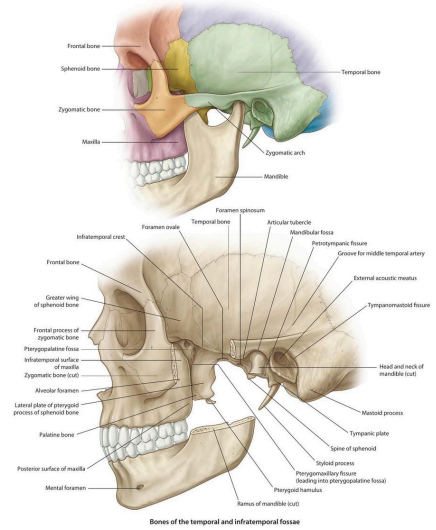
Infratemporal fossa after removing the ramus

Dr. Sameh Doss handout, p. 54



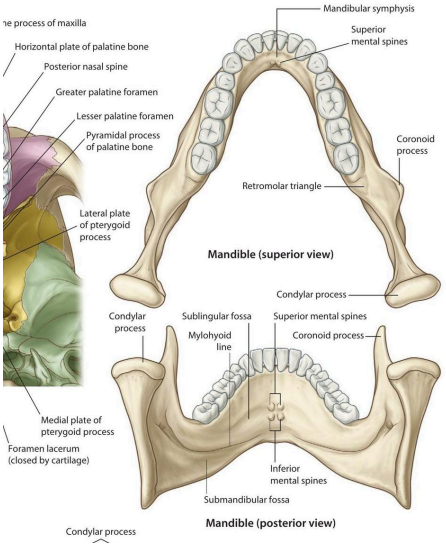
Temporal and infratemporal fossae: surface projection

Gray's Atlas of Anatomy, 2nd ed., p. 526



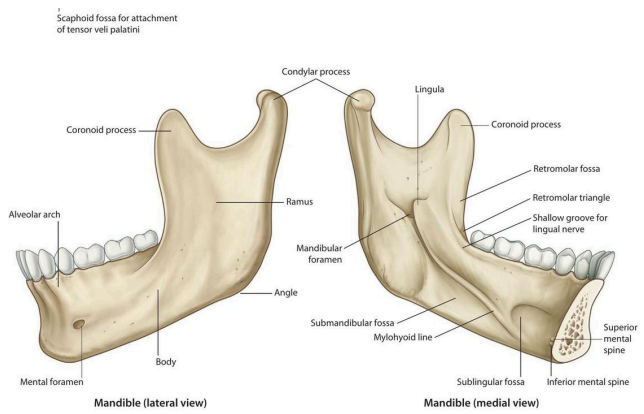
Bones of the temporal and infratemporal fossae

Gray's Atlas of Anatomy, 2nd ed., p. 527



Mandible (medial and posterior views)

Gray's Atlas of Anatomy, 2nd ed., p. 572



Mandible (lateral view)

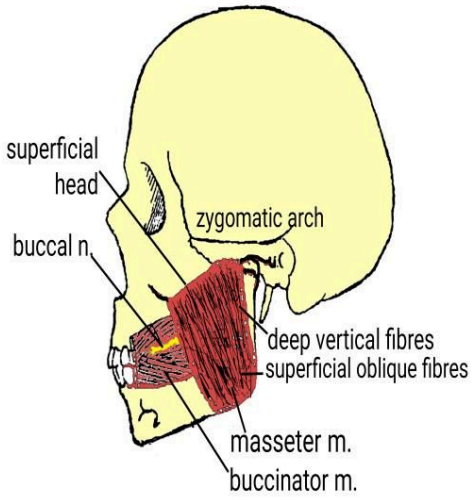
Gray's Atlas of Anatomy, 2nd ed., p. 572

02 Muscles of mastication

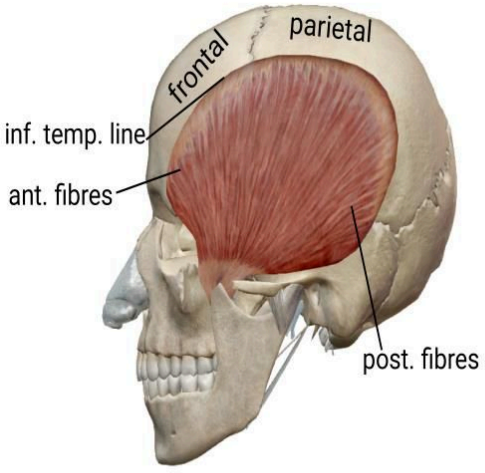
- All from the **1st arch** and supplied by **V3**.

- **Masseter:** zygomatic arch → lateral ramus. Elevates; superficial fibres protrude.
- **Temporalis:** temporal fossa → coronoid process. Elevates; **posterior fibres retract**.
- **Medial pterygoid:** deep head from the medial surface of the lateral pterygoid plate, superficial head from the maxillary tuberosity → medial angle of mandible. Elevates, protrudes, side to side.
- **Lateral pterygoid:** upper head (infratemporal surface of greater wing) → disc and capsule; lower head (lateral surface of lateral pterygoid plate) → pterygoid fovea of the neck. **The only muscle that opens the mouth** (with gravity and the suprahyoids); protrudes and moves the jaw to the opposite side.

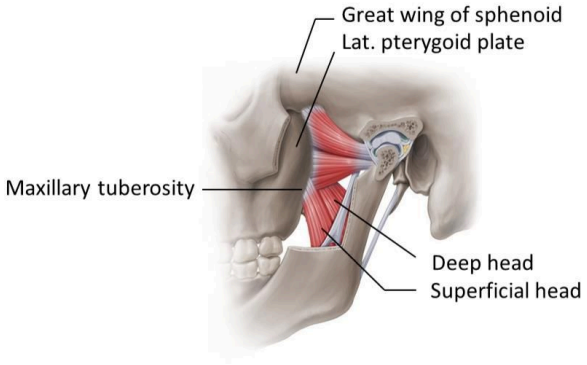
CLINICAL PEARL Masseter, temporalis and medial pterygoid close the jaw; lateral pterygoid opens it. Trismus usually reflects spasm or invasion of the medial pterygoid.



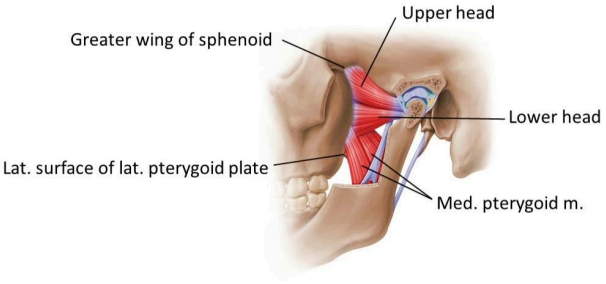
Masseter: superficial and deep heads
Dr. Sameh Doss handout, p. 51



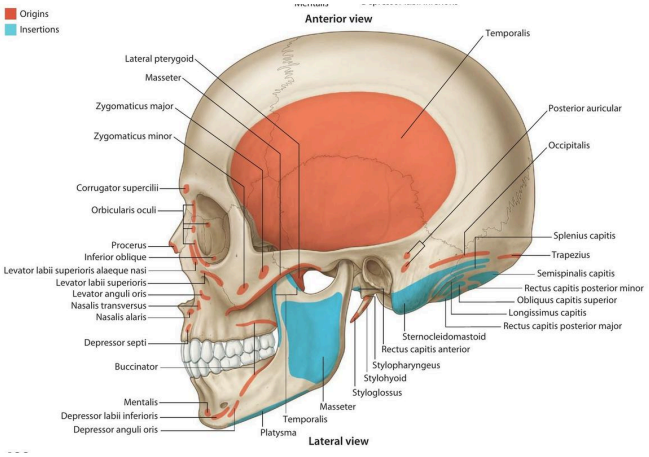
Temporalis
Dr. Sameh Doss handout, p. 52



Lateral pterygoid: upper and lower heads
Dr. Sameh Doss handout, p. 53

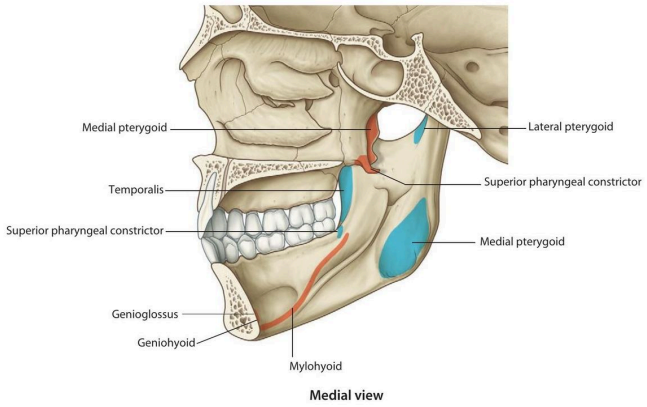


Medial pterygoid: deep and superficial heads
Dr. Sameh Doss handout, p. 53



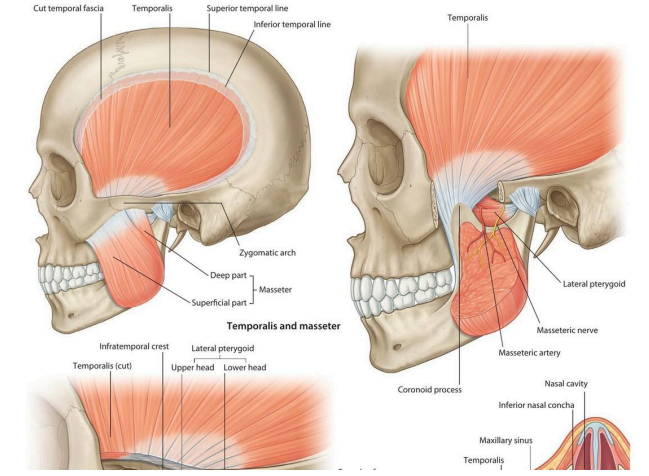
Skull, lateral view: muscle attachments

Gray's Atlas of Anatomy, 2nd ed., p. 488



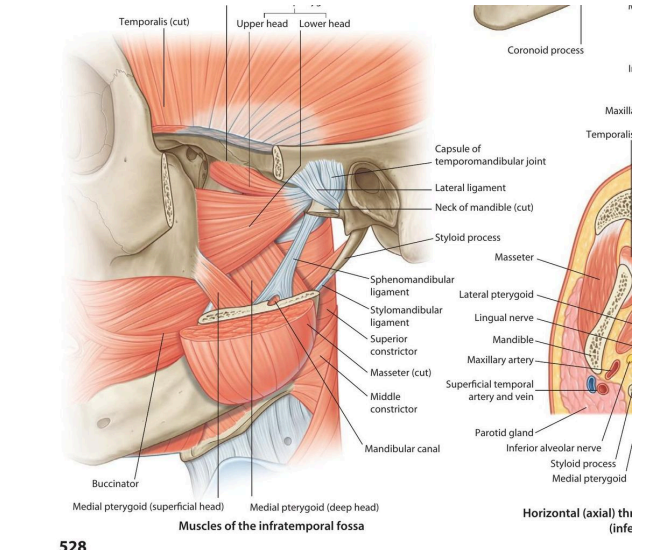
Mandible, medial view: pterygoid and constrictor attachments

Gray's Atlas of Anatomy, 2nd ed., p. 489



Temporalis and masseter

Gray's Atlas of Anatomy, 2nd ed., p. 528



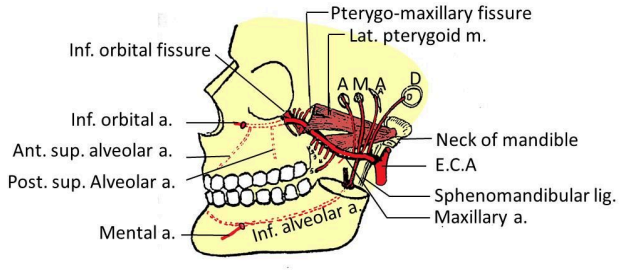
Muscles of the infratemporal fossa, with axial section

Gray's Atlas of Anatomy, 2nd ed., p. 528

03 Maxillary artery

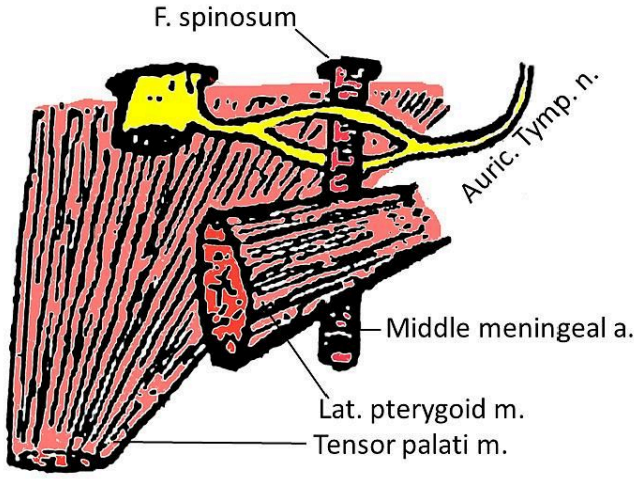
- Larger terminal branch of the ECA, from behind the neck of the mandible in the parotid.
- **1st (mandibular) part**, behind the neck of mandible: deep auricular, **anterior tympanic**, **middle meningeal** (through foramen spinosum), accessory meningeal, **inferior alveolar**.
- **2nd (pterygoid) part**, on or deep to the lateral pterygoid: deep temporal, pterygoid, masseteric, buccal branches (muscular).
- **3rd (pterygopalatine) part**, in the pterygopalatine fossa: posterior superior alveolar, infraorbital, descending (greater and lesser) palatine, artery of the pterygoid canal, pharyngeal, **sphenopalatine** (terminal, to the nose).

> **CLINICAL PEARL** The sphenopalatine artery leaves the fossa through the sphenopalatine foramen behind the middle turbinate: the target in endoscopic ligation for posterior epistaxis.



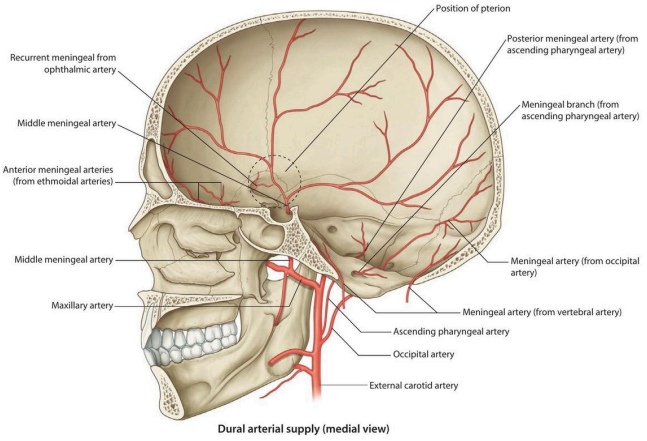
Maxillary artery and its three parts

Dr. Sameh Doss handout, p. 64



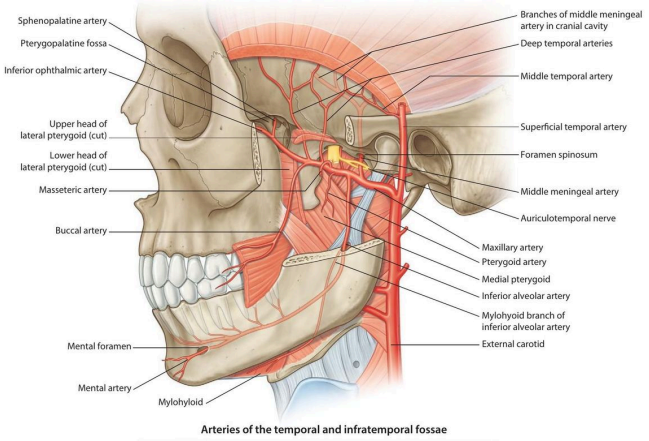
Relations of the lateral pterygoid: maxillary artery and V3 branches

Dr. Sameh Doss handout, p. 65



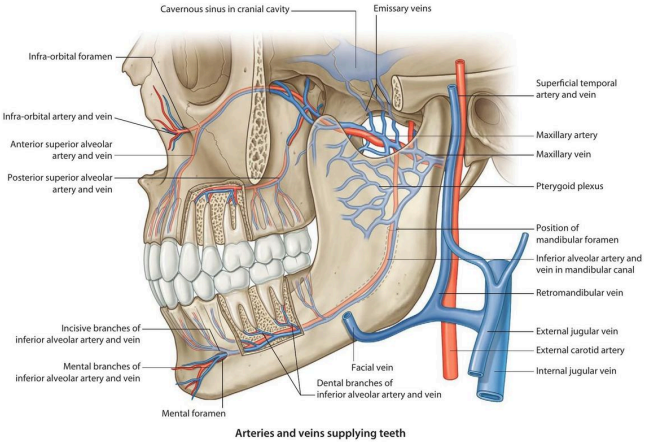
Dural arterial supply: middle meningeal artery (medial view)

Gray's Atlas of Anatomy, 2nd ed., p. 492



Arteries of the temporal and infratemporal fossae: maxillary artery

Gray's Atlas of Anatomy, 2nd ed., p. 533



Arteries and veins supplying the teeth

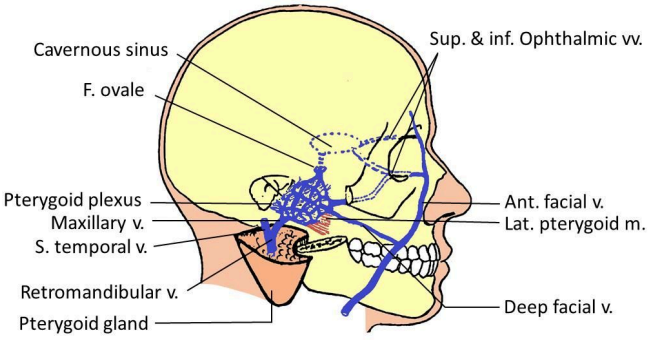
Gray's Atlas of Anatomy, 2nd ed., p. 576

04 Pterygoid venous plexus

- Around the lateral pterygoid; drains to the **maxillary vein** and then the retromandibular vein.

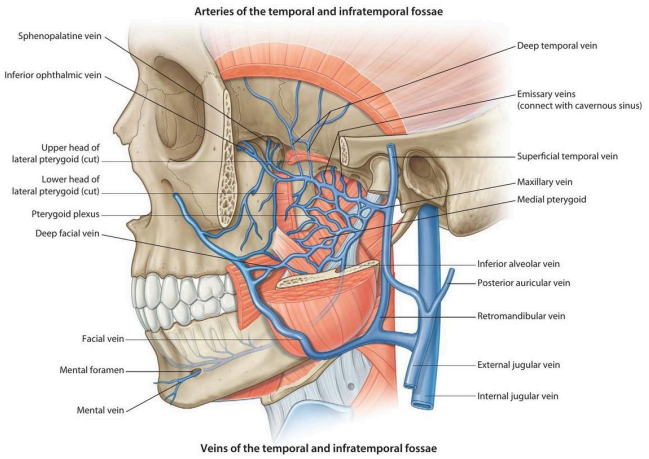
- **Connections:** cavernous sinus (emissary veins through foramen ovale and lacerum), inferior ophthalmic vein (inferior orbital fissure), facial vein (deep facial vein).

CLINICAL PEARL A route for spread of infection from the face, teeth and nose to the cavernous sinus.



Pterygoid venous plexus and its connections

Dr. Sameh Doss handout, p. 68



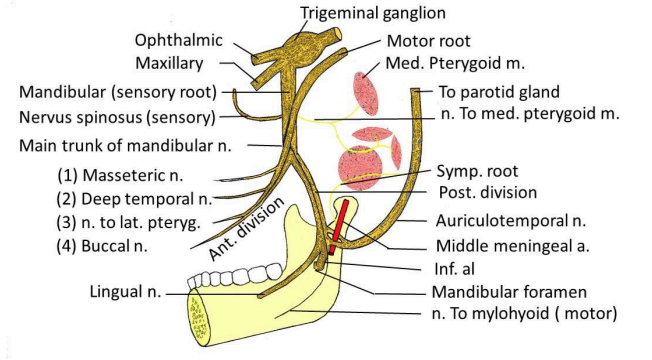
Veins of the temporal and infratemporal fossae: pterygoid plexus

Gray's Atlas of Anatomy, 2nd ed., p. 533

05 Mandibular nerve in the fossa

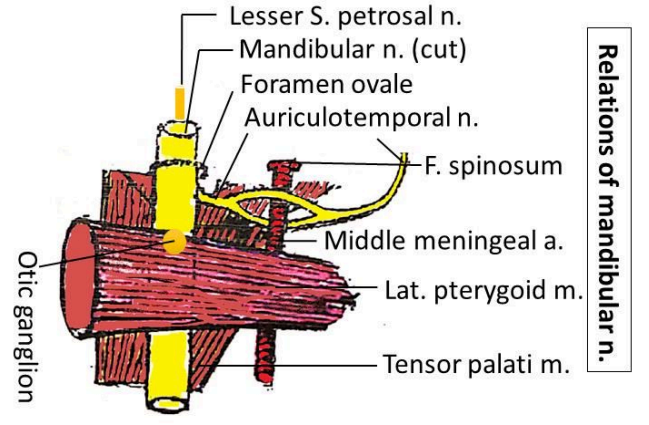
- Sensory and motor roots join in the **foramen ovale**; the trunk lies between the lateral pterygoid (lateral) and tensor veli palatini (medial), with the **otic ganglion** on its medial side.
- **Trunk:** nervus spinosus (meninges), nerve to medial pterygoid (also to tensor tympani and tensor veli palatini through the otic ganglion).
- **Anterior division (mainly motor):** masseteric, deep temporal, nerve to lateral pterygoid, and the **buccal nerve** (sensory, cheek).
- **Posterior division (mainly sensory): auriculotemporal** (encircles the middle meningeal artery), **lingual**, **inferior alveolar** (with nerve to mylohyoid, its only motor branch).
- **Chorda tympani** joins the lingual nerve deep to the lateral pterygoid.

SOURCES DIFFER ROOTS OF V3: Doss says they unite in the foramen ovale; Gray's says just below it in the infratemporal fossa.



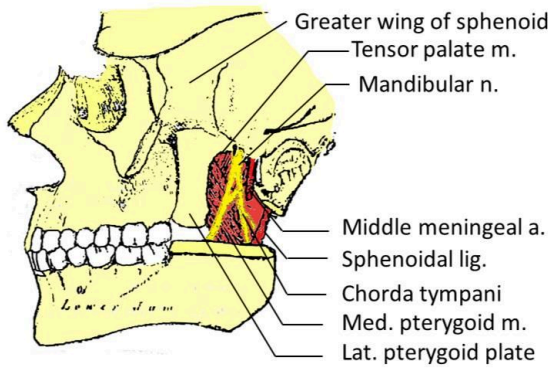
Mandibular nerve: trunk, divisions and branches

Dr. Sameh Doss handout, p. 55



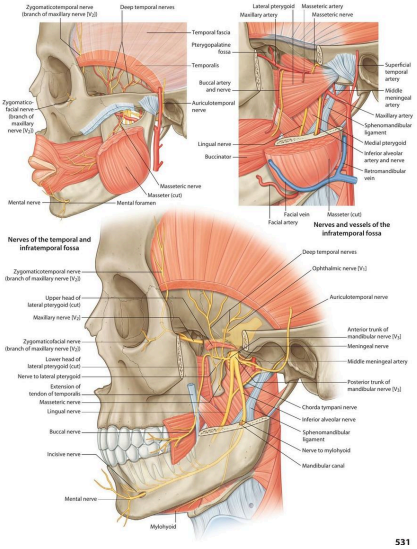
Relations of the mandibular nerve below the foramen ovale

Dr. Sameh Doss handout, p. 55



Deep infratemporal fossa: mandibular nerve and middle meningeal artery

Dr. Sameh Doss handout, p. 54



Nerves and vessels of the temporal and infratemporal fossae

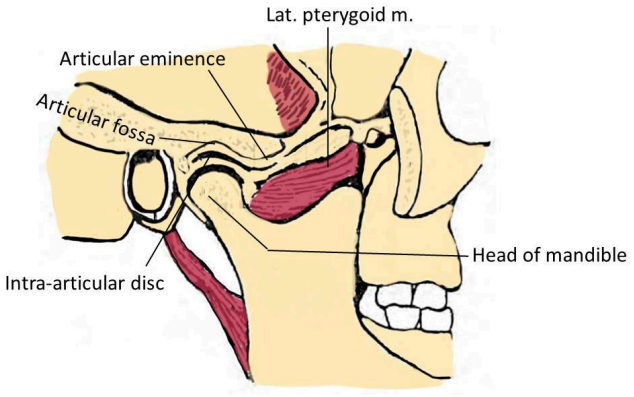
Gray's Atlas of Anatomy, 2nd ed., p. 531

06 Temporomandibular joint

- **Synovial, condylar** joint between the **mandibular fossa and articular eminence** above and the **head of mandible** below.
- A fibrocartilaginous **articular disc** divides it into **upper (gliding)** and **lower (hinge)** compartments; the lateral pterygoid inserts into the disc.
- **Ligaments:** lateral (temporomandibular) ligament is the main one; accessory ligaments are sphenomandibular, stylomandibular and pterygomandibular.
- **Nerves:** auriculotemporal and masseteric.
- **Opening:** first hinge in the lower compartment, then the condyle and disc glide forward onto the eminence in the upper compartment.
- The articular surfaces are covered with fibrocartilage, not hyaline cartilage (developed from membrane bone).

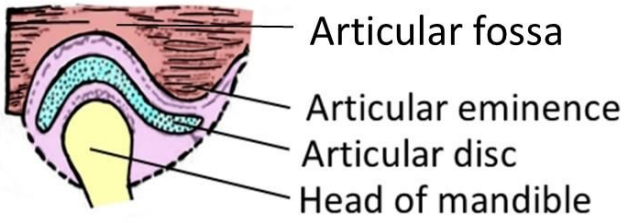
CLINICAL PEARL TMJ pain is a common cause of referred otalgia (auriculotemporal nerve). Anterior dislocation traps the condyle in front of the eminence; reduce by pressing down and back on the molars.

SOURCES DIFFER PTERYGOMANDIBULAR LIGAMENT: Doss lists it as an accessory ligament of the TMJ; Gray's lists only the sphenomandibular and stylomandibular.



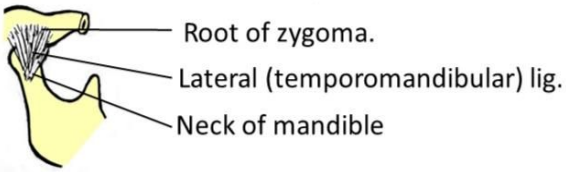
Temporomandibular joint: articular disc and lateral pterygoid

Dr. Sameh Doss handout, p. 199



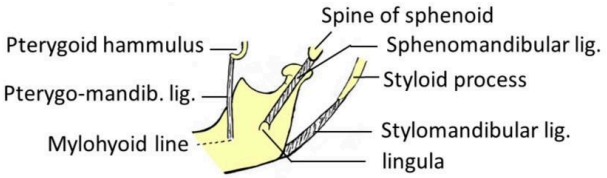
Compartments of the TMJ divided by the disc

Dr. Sameh Doss handout, p. 199



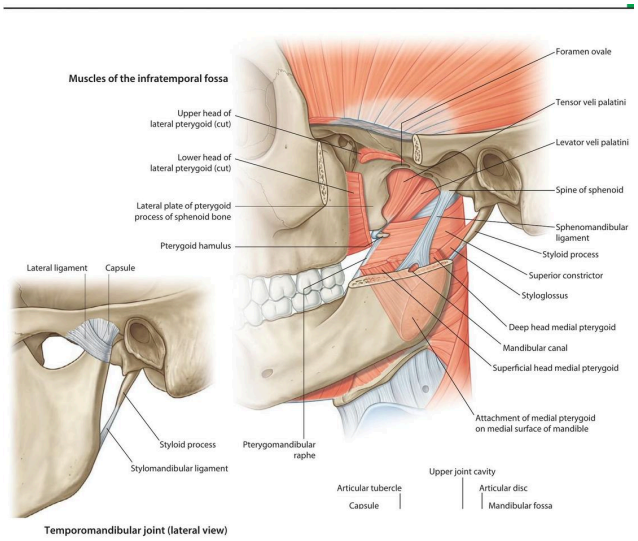
Lateral (temporomandibular) ligament

Dr. Sameh Doss handout, p. 199



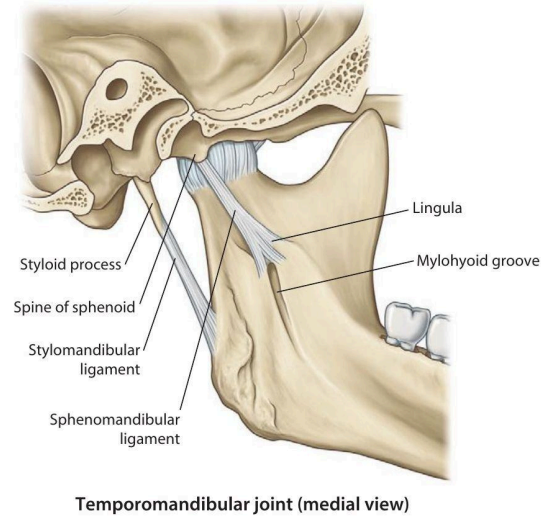
Accessory ligaments of the TMJ

Dr. Sameh Doss handout, p. 199



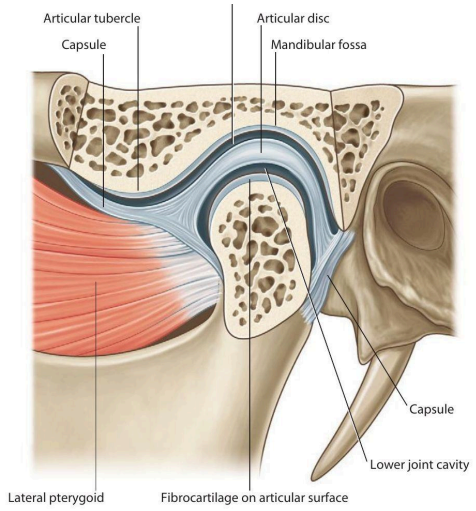
Temporomandibular joint (lateral view) and muscles of the infratemporal fossa

Gray's Atlas of Anatomy, 2nd ed., p. 529



Temporomandibular joint (medial view)

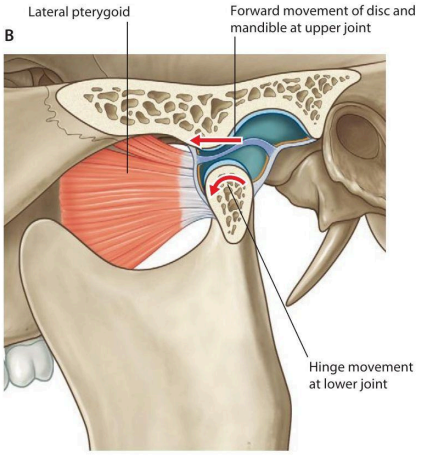
Gray's Atlas of Anatomy, 2nd ed., p. 529



Sagittal section through temporomandibular joint

Sagittal section through the temporomandibular joint

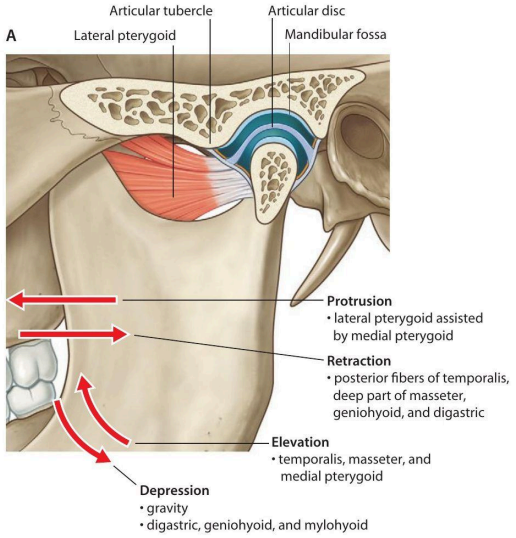
Gray's Atlas of Anatomy, 2nd ed., p. 529



Movements of the temporomandibular joint
A. Mouth closed B. Mouth open

Movements of the temporomandibular joint

Gray's Atlas of Anatomy, 2nd ed., p. 530

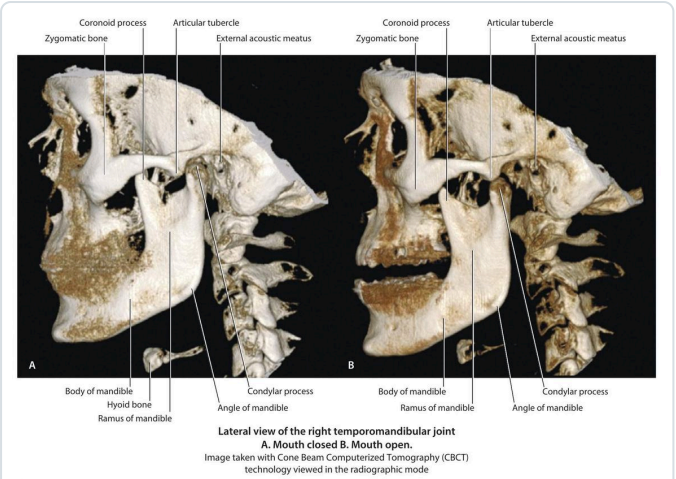


Muscles producing protrusion, retraction, elevation and depression

Gray's Atlas of Anatomy, 2nd ed., p. 530

07 Radiological anatomy: On imaging

- Fat planes in the infratemporal fossa and around the pterygoid muscles are lost early with tumour invasion; **MRI** shows perineural spread along V3 through the foramen ovale (enlarged foramen, enhancement).
- **TMJ MRI** shows disc position (anterior displacement with or without reduction).



Right TMJ, mouth closed and open: cone-beam CT

Gray's Atlas of Anatomy, 2nd ed., p. 530

08 Surgical anatomy: In theatre

- **Approaches:** transmandibular, postauricular infratemporal fossa approach (Fisch types A–C), preauricular subtemporal–infratemporal approach, transmaxillary, and endoscopic transpterygoid through the back wall of the maxillary sinus.
- Ligate the **maxillary artery** early when resecting tumours here; the pterygoid plexus bleeds freely.

T Tables

Branches of the maxillary artery

Part	Position	Branches
1st (mandibular)	Behind the neck of mandible	Deep auricular, anterior tympanic, middle meningeal, accessory meningeal, inferior alveolar
2nd (pterygoid)	On or deep to lateral pterygoid	Masseteric, deep temporal, pterygoid, buccal
3rd (pterygopalatine)	In the pterygopalatine fossa	Posterior superior alveolar, infraorbital, descending palatine, artery of pterygoid canal, pharyngeal, sphenopalatine

Muscles of mastication

Muscle	Origin	Insertion	Main action
Masseter	Zygomatic arch	Lateral ramus and angle	Elevates; superficial fibres protrude
Temporalis	Temporal fossa and fascia	Coronoid process and anterior ramus	Elevates; posterior fibres retract
Medial pterygoid	Medial surface of lateral pterygoid plate; maxillary tuberosity	Medial angle of mandible	Elevates, protrudes, side to side
Lateral pterygoid	Greater wing (upper head); lateral pterygoid plate (lower head)	Disc and capsule; pterygoid fovea of neck	Opens mouth, protrudes, moves jaw to opposite side

All supplied by V3; all from the 1st arch.

Movements of the TMJ

Movement	Muscles
Depression (opening)	Lateral pterygoid, digastric, mylohyoid, geniohyoid, gravity
Elevation (closing)	Masseter, temporalis, medial pterygoid
Protrusion	Lateral and medial pterygoids, superficial masseter
Retraction	Posterior fibres of temporalis
Side to side	Medial and lateral pterygoids of alternate sides

M Mnemonics

Muscles of mastication

Only the lateral pterygoid opens

Masseter, temporalis and medial pterygoid close the jaw

Maxillary artery, 1st part

DAMAI

Deep auricular, Anterior tympanic, Middle meningeal, Accessory meningeal, Inferior alveolar

Foramen spinosum

Spinosum = Spinal tap of the meninges

Middle meningeal artery and nervus spinosus

Q Rapid recall

1. **Roof of the infratemporal fossa?** Infratemporal surface of the greater wing of sphenoid with foramina ovale and spinosum.
2. **Contents of the infratemporal fossa?** Pterygoid muscles, maxillary artery (1st and 2nd parts), pterygoid plexus, V3 and branches, chorda tympani, otic ganglion, sphenomandibular ligament.
3. **Only muscle that actively opens the mouth?** Lateral pterygoid (with the suprahyoid muscles and gravity).
4. **Which part of temporalis retracts the mandible?** Posterior (horizontal) fibres.
5. **Nerve supply of the muscles of mastication?** Mandibular nerve (V3).
6. **Branches of the first part of the maxillary artery?** Deep auricular, anterior tympanic, middle meningeal, accessory meningeal, inferior alveolar.
7. **Terminal branch of the maxillary artery?** Sphenopalatine artery.
8. **Main ligament of the TMJ?** Lateral (temporomandibular) ligament.
9. **Which TMJ compartment allows hinge movement?** The lower compartment (between disc and condyle).
10. **Nerve supply of the TMJ?** Auriculotemporal and masseteric nerves.
11. **Connections of the pterygoid venous plexus?** Cavernous sinus (emissary veins), inferior ophthalmic vein, facial vein (deep facial), maxillary vein.
12. **Accessory ligaments of the TMJ?** Sphenomandibular, stylomandibular and pterygomandibular.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.